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L2	inez\n\nThe	\xe9n-1-	Afee and the other	er the left of	Sons\n\nThe
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	\n de la c	ring and the other	Leanmig.	er \ufffd\ufffdI	Oklahoma first time I
	\n de la c	ring and the next	Afee\n\nThe	world and the future	Austria book.\n
	\n de la c	ring and the other	Leanaway from the	mer 1, 1	Australia movie.\n
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	\n, and the	ellLean, and	Lean\n\nThe	walker be the first	Australia movie "The
	\n, and the	ellDonough,	Lean\n\nThe	te Marty McFly	Australia movie "The
	\n"\n	ellDonough,	Lean\n\nThe	te Marty McFly	Vietnam movie "The
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	\n" id="	GreenbergDonough,	Lean\n\nThe	movies Marty McFly	Boston movie "Back
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	\npng" alt	Mc he was a	Fly\n\nThe	arrives Marty McFly	1984 Back to the
	\npng" alt	Mc and I'm	Fly\n\nThe	arrives Marty McFly	1989 to the Future
	\npng" alt	Mc and I'm	Fly\n\nThe	's\nThe first	Back to the Future
L26	\n1.0	Mc and I'm	Fly\n\nThe	(\nThe first	Back movie "Back
L27	.\n\nThe	Mc and the other	Fly.\n\n	,\nThe first	the movie "Back
L28	y\n\nThe	Mc and I'm	Fly.\n\n	\n and the future	Back future.\n

Figure 6: The Future Lens applied to the hidden states of GPT-J-6B processing *Marty McFly from*. Each cell illustrates the most likely sequence of future tokens that the respective hidden state predicts. The darker boxes correspond to higher probabilities/confidence.

et al. 2023). Other work (Haviv et al., 2023) has shown that earlier layers of models tend to promote memorized concepts or tokens, while later layers boost model confidence in these tokens. Our paper can be viewed as an extension of this work on investigating memorization of multi-token phrases: we ask whether and to what extent a single model hidden state encodes multi-token information.

Prompt Tuning Prompt Tuning has emerged as a parameter-efficient method for fitting LMs for new downstream tasks. By freezing the LM and optimizing only the soft prompt parameters, models are able to achieve performance comparableto that observed after fine-tuning all parameters. Li *et al.* (2021) introduced prefix tuning which entailed training plug-and-play prefix that steers the behavior of the LMs for the downstream tasks. Other work (Wallace et al., 2019) applied a gradient-based method to search for the best discrete prompts which enable the model to produce desire generation. Sun and colleagues (2023) train the prefix soft prompt as a way of aligning semantically equivalent instructions in latent space.

5 Discussion

In this paper, we explored the degree to which we are able to decode multi-token outputs subsequent to a particular token on the basis of its hidden representation alone. The results in Table 2 and Figures 4 and 5 indicate that the representations encode such information, to some degree. Among the decoding methods we assessed, learned prompts are best able to predict such future tokens. Both the linear and the learned prompt models achieve better accuracy than the empirical bigram baseline at $N = 1$. Interestingly, predictive accuracy of the learned prompt model peaks at the middle-layer hidden states, suggesting that subsequent-token information is encoded at those layers; this pattern is very different from the immediate next-token $N = 0$, in which accuracy peaks at the last layer.

The learned prompt model realizes an accuracy sufficient to be potentially useful as a "Future Lens" to provide insights about subsequent token information contained in hidden states within LLMs. This provides a way to decode a short sequence of tokens encoded in a hidden state, rather than only the single immediate token prediction.

Data and Code Availability All code and data for demo and implementation is made available at: https://future.baulab.info

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A Appendix

Additional Figures

In this main paper, we report results based on models that are trained to optimize the $N = 1$ single token-ahead prediction, and we test those models for predictive accuracy for other N.

The same methods can also be used to optimize subsequent tokens, and the results of those methods are shown here. We find that optimizing for $N = 1$ works best and generalizes surprisingly well to other N, but that that optimizing for other N does not perform well for $N = 1$.

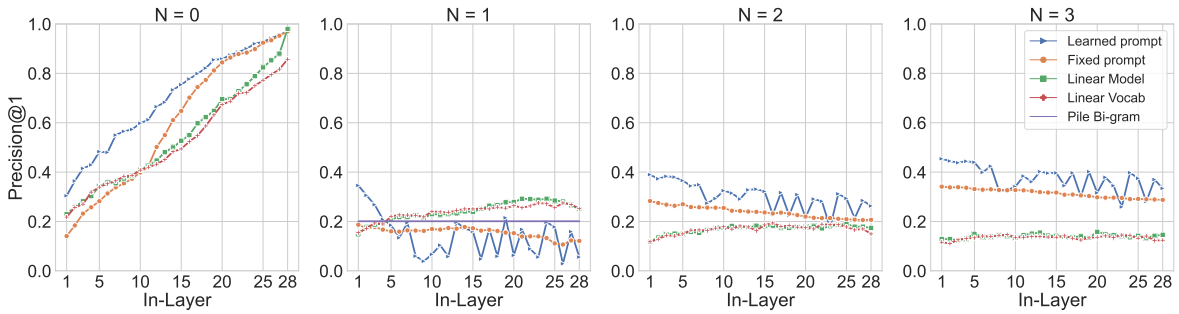


Figure 7: The Precision@1 (Accuracy) of all the methods trained with predicting the currently decoded token (teacher-forcing)

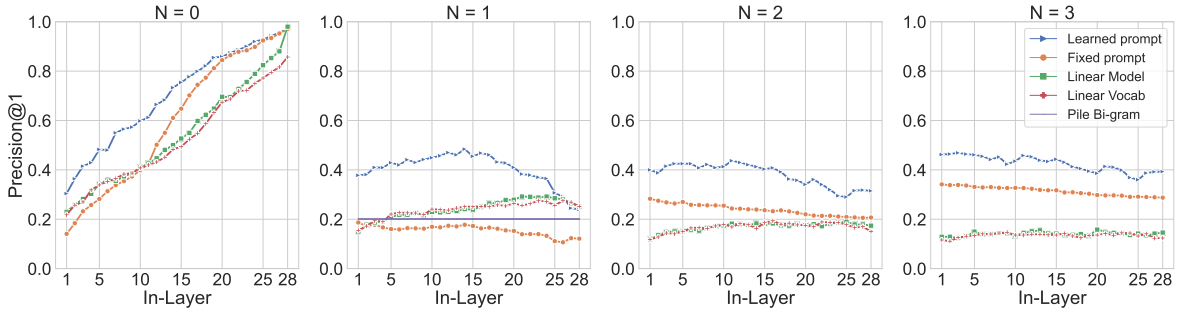


Figure 8: The Precision@1 (Accuracy) of all the methods trained with predicting the 1st next token

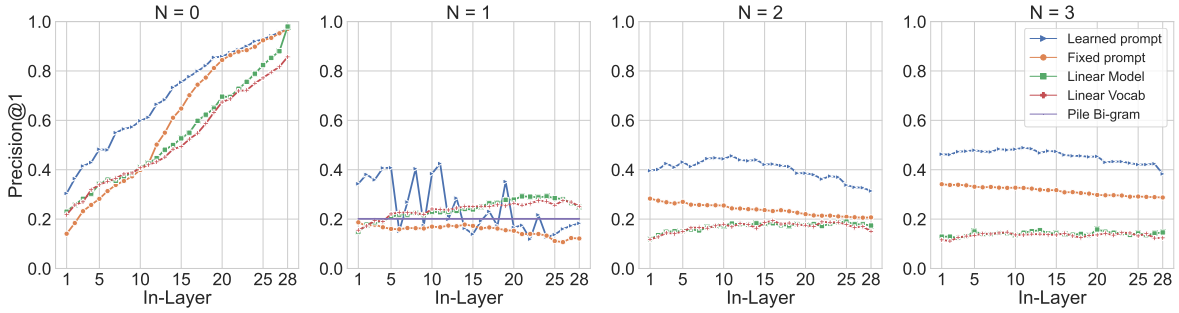


Figure 9: The Precision@1 (Accuracy) of all the methods trained with predicting the 2nd next token